

App Development Project Report

App Dev Project Report

1. Student Details

Name: Samreen Kausar

Roll Number: 24F3000718

Email: 24F3000718@onlinedegree.iitm.ac.in

About Me: I am a Course Student at IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Placement Portal

Problem Statement:

Institutes rely on spreadsheets, emails, and manual processes to manage campus recruitment activities. This makes it difficult to coordinate company approvals, track student applications, prevent duplicate registrations, and maintain historical placement records across multiple stakeholders.

Approach:

We built a role-based web application using Flask that separates concerns across three user roles — Admin, Company, and Student. Each role has its own dashboard and is restricted to only the actions relevant to their function.

The workflow is: Company registers → Admin approves → Company creates drives → Admin approves drives → Students apply → Company shortlists/selects.

3. AI/LLM Declaration

I used **Claude (by Anthropic)** to assist in the development of this project.

The AI was used for approximately **20-25%** of the project, specifically for:

Generating Flask route structures and Jinja2 template syntax.

Identifying SQL query syntax errors and parameter issues.

Designing the dashboard layouts.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework routing,templating sessions
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap	Frontend styling and responsive design
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data

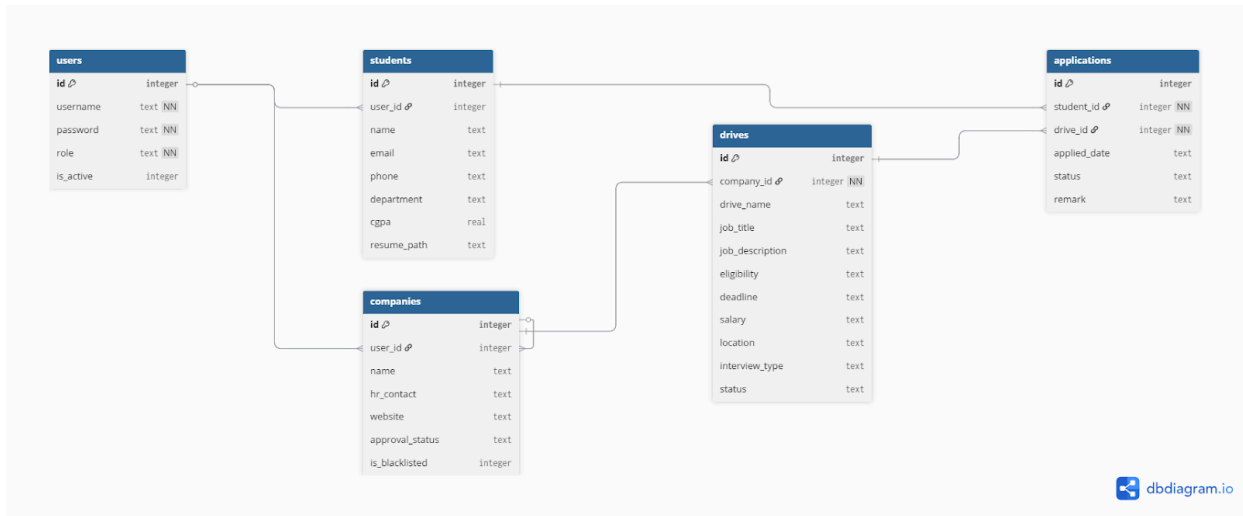
5. Database Schema / ER Diagram

Tables:

1. **Users** — stores user profile details (id,username , password , role , is_active)
2. **Students**— Stores student profiles linked to users (id, user_id, name, email, cgpa, resume_path , is_blacklisted).
3. **Companies:** Stores company profiles (id, user_id, name, hr_contact, status).
4. **Drives:** Logs recruitment events (id, company_id, job_title, eligibility, status).
5. **Applications:** Tracks student applications (id, student_id, drive_id, status, remark).

Relationships:

- **One-to-One:** User → Student and User → Company.
- **One-to-Many:** Company → Drives.
- **One-to-Many:** Student → Applications and Drives → Applications.
- **Constraint:** A **UNIQUE** constraint is applied to (student_id, drive_id) to prevent duplicate applications



6. API Resource Endpoints

Endpoint	Method	Description
/login	GET / POST	Authenticate user and generate session
/register/student	GET / POST	Creates new user and student profile rows
/register/company	GET / POST	Company registration
/logout	GET	Clears session and redirects to login

7. Architecture and Features (optional)

Architecture Overview:

Architecture Overview

- **app.py:** Main entry point for the Flask application.
- **Database:** Single SQLite file for local portability.

Implemented Features:

Role-Based Access Control: Separate dashboards for Admin, Company, and Student.

Workflow Automation: Company blacklisting automatically closes active drives.

Data Integrity: Prevention of duplicate applications at the database level.

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1KZ79GY03ODJf5nV66QvHZp1dV7zx5z4p/view?usp=drive_link

 mad -2 video.mp4
